HINTS & SOLUTION

Q.1 (C)

Use $P_A = C_ART$ and $C_A = \frac{n_A}{V}$

Q.2 (C)

Q.3 (B)

Overall order of photochemical reactions is not always zero.

Q.4 (A)

Q.5 (B)

	A	+	B	$\longrightarrow$	C	+	D
t = 0	1		1				
't'	1-x		1-x		x		x

rate = k[A]½ [B]½

$-\frac{d[A]}{dt} = k(1-x)^{1/2}(1-x)^{1/2} \quad \Rightarrow \quad -\frac{d(1-x)}{dt} = k(1-x) \quad \Rightarrow \quad \frac{dx}{dt} = k(1-x)$

$\Rightarrow \int_0^x \frac{dx}{1-x} = k\int_0^t dt \quad \Rightarrow \quad [-\ln(1-x)]_0^x = kt \quad \Rightarrow \quad -\ln(1-x) = kt$

$\Rightarrow -\ln(1-x) = kt \quad \Rightarrow \quad t = \frac{1}{k}\ln\left(\frac{1}{1-x}\right)$

When [A] = 0.25

$\therefore$ x = 0.75

$\therefore t = \frac{1}{2.31\times10^{-3}}\ln\frac{1}{0.25} = \frac{10^3}{2.31}\ln 4 = \frac{2\times10^3\times0.693}{2.31} = \frac{2\times10^3\times0.693}{2.31} = 600$ Sec.]

Q.6 (C)

Q.7 (C)

Q.8 (D)

Q.9 (C)

$A_{(aq)} + H_2O \rightarrow B_{(aq)} + C_{(aq)}$

Rate = K[A] [H_2]

Reaction is pseudo first order reaction w.r.t. A

$\therefore$ Rate = K' [A]

where K' = K[H_2O]

$$K' = \frac{2.303}{6.93} \log \frac{1}{1/4} = \frac{1}{5}$$

Also [H_2O] = 55.5

$$\therefore K = \frac{K'}{[H_2O]} = \frac{1}{5 \times 55.5} = 3.6 \times 10^{-3}$$

Q.10 (B)

$K = Ae^{-Ea/RT}$

$$= 1.25 \times 10^8 \ e^{-\frac{16}{2\times10^{-3}\times400}} = 1.25 \times 10^8 \ e^{-20}$$

$$= \frac{1.25 \times 10^8}{5 \times 10^8} = \frac{1}{4} \ sec^{-1}$$

$$t_{avg} = \frac{1}{K} = 4 \ sec]$$

Q.11 (A)

Q.12 (C)

$$\text{Rate law} = \frac{dp}{dt} = K_2 [R] [C]$$

$$\text{at steady state } \frac{d[R]}{dt} = 0$$

$$\frac{d[R]}{dt} = K_1 [A] - K_2 [R] [B] - K_2[R] [C] = 0$$

$K_1 [A] = \{K_2[B] + K_2[C]\} [R]$

$$[R] = \frac{K_1[A]}{K_2\{[B]+[C]\}}$$

$$\text{Rate} = K_2 \times \frac{K_1}{K_2} \frac{[A]}{[B]+[C]} \times [C]$$

$$= \frac{K_1[A]}{\frac{[B]}{[C]} + 1}$$

where C is very high

Rate = K_1 [A]

$\therefore$ order of reaction = 1

Q.13 (C)

$$K = A\,e^{-\frac{Ea}{RT}}$$

$$\ln K = \ln A - \frac{Ea}{RT}$$

Q.14 (A)

At Steady State

$K_1[A] = K_2[B]$ and $K_3[C] = K_2\,[B]$

$\therefore$ $K_1\,[A] = K_3\,[C]$

$$\frac{[A]}{[C]} = \frac{K_3}{K_1} = 40000$$

Q.15 (B)

Q.16 (A)

Q.17 (C)

(i) $CH_4 + {\bullet}OH \rightarrow CH_3^{\bullet} + H_2O$

$r = k[CH_4][{\bullet}OH]$ where $k = 4 \times 10^6$ and $[{\bullet}OH]$ constant

$\therefore$ Reaction is pseudo Ist order

$r = k'\,[CH_4]$

$k' = k\,[{\bullet}OH]$

$= 4 \times 10^6 \times 1 \times 10^{-15} = 4 \times 10^{-9}$

$$\therefore\ t_{av} = \frac{1}{k'} = \frac{1}{4\times10^{-9}} = 25\times10^{7}\ \text{sec.}$$

(ii&iii) t_{av} of CH_4 is 25×10^7 sec.

and t_{av} of isoprene $= \dfrac{1}{2\times10^{11}\times10^{-15}} = 4000$ sec.

t_{av} of CH_4 is much more than isoprene, there for it will travel longer distance as well as remain for longer time in atmosphere.

Q.18 (A) (B) (D)

(A) $\Delta H = [2(-1263)] - [(-2238) + (-285)] = -3$ kJ

(B) $C_{12}H_{22}O_{11} + 12O_2 \longrightarrow 12\,CO_2 + 11\,H_2O$

$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O \times 2$

$C_{12}H_{22}O_{11} + H_2O \longrightarrow 2C_6H_{12}O_6\ ;\ \Delta H = -3$

Hence, $\Delta_C\ H_{C_{12}H_{22}O_{11}}$ is greater than $2 \times \Delta_C\ H_{C_6H_{12}O_6}$

(C) Hydrolysis is exothermic $T\uparrow h\downarrow$

(D) Concentration becomes one-forth is each equal time internal Hence, 1st order

Q.19 (A) (D)

Q.20 (A) P (B) Q,R (C) S (D) T

Q.21 1202 or 0240

Reaction is of 2nd order because $t_{1/2} \alpha \dfrac{1}{a}$

initial $t_{1/2} = 120$ min.

Q.22 (a) 1, (b) 6.93×10^{-3} min^{-1} , (c) 200, (d) 950 mm

Q.23 $(0.4)^{3/2} = 0.253$

Q.24 0.0805

Q.26 (a) 1, (b) 6.93×10^{-3} min^{-1} , (c) 200, (d) 950 mm

Q.27 0050.00
Since B is in large excess, rate law will become
Rate = $k'[A]^2$ where $k' = k[B]$

$$\frac{1}{[A]_t} - \frac{1}{[A]_0} = k't$$

$$\frac{1}{[A]_0 k'} = t_{1/2}$$

$$t_{1/2} = \frac{1}{0.2 \times 5 \times 10^{-5} \times 2 \times 10^3} = 50 \text{ minutes}$$

Q.28 2

$$SO_2Cl_2(g) \longrightarrow SO_2(g) + Cl_2(g)$$

10 mmole

10–x x x

SO_2 + Hypo = I_2
x × 2 + 1 × 10 = 1 × 25
x = 7.5

Q.29 0006

$$\frac{K_{500}}{K'_{500}} = e^{\frac{693}{1000}}$$

$$K'_{500} = K_{500} . e^{\frac{693}{1000}} = K_{500} \times e^{.693} \qquad [\because e^{0.693} = e^{\ln 2}]$$

$$K'_{500} = 2K_{500}$$

$$K_{500} = \frac{\ell n2}{6}$$

For 75% reaction

$$K'_{500} \times t = \ell n4$$

t = 6 min.

Q.30 4

$$\left[\begin{array}{l}\text{For } n^{th} \text{ order } (n > 1) \text{ reaction} \\ \dfrac{1}{[A]^{n-1}} \text{ vs time graph is a straight line always.}\end{array}\right]$$

$\dfrac{1}{[A]^4}$ vs time graph is given straight line

$\therefore n - 1 = 4$

$n = 5$

Reaction is of 5th order

$$\therefore -\frac{dA}{dt} = K_A[A]^5$$

$$\frac{dA}{dt} = K_A[A]^5$$

$$\int_{[A]_0}^{[A]} \frac{d[A]}{[A]^5} = -k_A \int_0^t dt$$

$$\Rightarrow -\frac{1}{4}\left[\frac{1}{[A]^4} - \frac{1}{[A]_0^4}\right] = -K_A t$$

$$\frac{1}{[A]^4} = \frac{1}{[A]_0^4} + 4K_A t$$

$\therefore 4K_A = \tan 45° = 1$

$K_A = 1/4 \qquad \therefore K_R = \dfrac{K_A}{2} = \dfrac{1}{8}$

$\therefore$ ROR $= K_R [A]^5 = \dfrac{1}{8} [2]^5 = 0004$ Ans.